Empirical OEV & Liquidation Dynamics: Protocol Value Recapture in Chainlink SVR and Structural Centralization on Ethereum L1

An Open Science Microstructure Audit Across 1,157 On-Chain Liquidations (Document ID: WERRACLE-RES-2026-001-REV2)

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ABSTRACT

Order Flow Auctions (OFA) and Proposer-Builder Separation (PBS) have fundamentally reshaped decentralized finance (DeFi) liquidation mechanics. We present an exhaustive empirical study of 1,157 verified on-chain liquidations encompassing 1,049 Ethereum L1 transactions and 108 multi-call decoded Chainlink Smart Value Recapture (SVR) transactions (105 on Base L2, 3 on Arbitrum) recorded during the acute volatility regime of October 12–19, 2026. Across 105 verified Base SVR transactions totaling $190,506.31 in gross liquidated collateral, we provide the first independent empirical proof that Chainlink SVR redirects 98.10% ($186,889.85 USD) of gross liquidation value directly back to lending protocols and depositors as competitive bids. Unconstrained Bertrand price competition among permissionless solvers compresses gross margins to 1.86% ($3,545.73 USD pre-swap), with median transaction margin settling at -1.55%. In 61.9% of transactions (65/105), solvers concluded at zero or negative net margin prior to off-chain DEX slippage. We identify active participation from 20 distinct solver contract architectures and 27 signing EOAs, debunking monopolistic capture (top solver captured only 31.64% volume share). Furthermore, we document that 64.76% (68/105) of SVR transactions were atomic multi-call bundles executing up to 32 sub-calls in a single transaction, explaining why naive single-event indexers severely misreport liquidation accounting. In contrast, analysis of 1,049 Ethereum L1 liquidations reveals an acute power-law distribution where 80.84% (848 txs) of block builder priority fees are micro-dust (< $1.00 USD, median $0.08 USD), while the top 22 whale liquidations capture 96.05% ($344,155.62 USD) of all builder revenue via private PBS bundles. All dataset records, replication scripts, and decoding tools are fully open-sourced in this Zenodo archive.

1. Introduction & Theoretical Background

In decentralized lending protocols (such as Morpho Blue and Aave v3), solvency is preserved via programmatic liquidations: when a borrower's Health Factor (HF) falls below 1.00, third-party liquidators seize collateral at a pre-configured bonus to repay outstanding debt. On transparent mempools, this design induces destructive Priority Gas Auctions (PGA) and Miner/Maximal Extractable Value (MEV) leakage, transferring surplus away from depositors to block builders. To mitigate this, Layer-2 rollups have introduced Order Flow Auctions (OFA), exemplified by Chainlink Smart Value Recapture (SVR) operating via Atlas metacalls. Despite significant theoretical interest, empirical measurement of SVR performance under severe market volatility has remained absent from peer-reviewed literature. This monograph provides the first machine-verified empirical accounting of SVR value recapture, solver competition, and multi-call bundling complexity.

2. Base L2: Chainlink SVR Financial Breakdown & Value Recapture

During the high-volatility window of October 12–19, 2026, 105 SVR liquidation events occurred on Base across 83 monitored Morpho Blue markets and Aave v3. Table 1 summarizes the macroeconomic value distribution across these 105 audited multi-call events.

Financial Metric	Amount (USD)	Share of Gross (%)	Microstructure Role
Gross Seized Collateral	$190,506.31	100.00%	Total borrower collateral liquidated
Protocol Recaptured Bids	**$186,889.85**	**98.10%**	Returned directly to protocol & depositors
Solver Retained Margin (Pre-Swap)	$3,545.73	1.86%	Pre-DEX slippage gross solver compensation
L2 Network Gas Fees	$70.73	0.04%	Base sequencer execution gas

Bertrand Margin Compression: Across the 105 transactions, solver margins exhibited severe downward compression. The median margin was -1.55%. Exactly 65 of 105 transactions (61.9%) concluded at zero or negative net profit prior to DEX pool swap slippage. Only 9 transactions (8.6%) realized margins ≥ 5.00%. This confirms that permissionless order flow auctions force searchers into intense Bertrand competition where bids approach the theoretical collateral ceiling.

3. Solver Decentralization & The Absence of Monopolistic Capture

A critical governance question is whether MEV order flow auctions concentrate into monopolistic cartels. We identified 20 unique solver contract architectures and 27 unique signing EOAs actively competing on-chain. Table 2 provides the volume and transaction distribution

among the top solvers.

Solver Contract Address	Tx Count	Gross Vol (USD)	Vol Share (%)	Net Pre-Swap PnL
0x8b61f5925f4515edb0b7b76c38036aaf9a818472	51 (48.6%)	$60,269.18	31.64%	+$447.20
0x3ae5122d62e78dc14b98d8d30aca4faf9d1f1bba	6 (5.7%)	$53,452.63	28.06%	+$1,968.45
0xbd0dd817c608e552d5f1ae701ab6c4da4fb00005	4 (3.8%)	$33,057.97	17.35%	+$620.40
0xd12810b19b596347a3afac206d3ca65d08594b3f	15 (14.3%)	$17,842.90	9.37%	+$182.15
0xa3eec55f71e62431b4afdb6b6734c11204f29a93	1 (1.0%)	$8,705.60	4.57%	+$212.80
Remaining 15 Solvers	28 (26.6%)	$17,178.03	9.01%	+$114.73

No Single-Entity Monopoly: Although solver `0x8b61...` won 48.6% of transaction counts by dominating micro-liquidations, it accounted for only 31.64% of total dollar volume. Large liquidations were vigorously contested by independent solvers `0x3ae5...` and `0xbd0d...`, driving bids up to 96%–98% on major positions.

4. Multi-Call Bundling Complexity & The Fallacy of Single-Event Indexers

A critical methodological contribution of this research is identifying the prevalence of atomic multi-call bundling. 64.76% (68 of 105 transactions) contained multiple liquidation sub-calls bundled into a single transaction execution. Single transactions executed up to 32 liquidation sub-calls atomically in a single block. The distribution was: 1 call (37 txs, 35.2%), 2 calls (14 txs, 13.3%), 3 calls (12 txs, 11.4%), 15 calls (8 txs, 7.6%), 4 calls (7 txs, 6.7%), 6 calls (5 txs, 4.8%). Naive indexers evaluating only the initial log receipt produce massive accounting errors (e.g. misreporting that solvers suffered -$58k losses). Accurate SVR measurement requires recursive sub-call transaction trace decoding.

5. Market Risk Distribution & Empirical Case Studies

Liquidation activity during the volatility window concentrated heavily in specific collateral assets across the 83 monitored Morpho Blue markets:
• cbXRP/USDC (LLTV 62.5%): 10 SVR transactions, $88,801.33 gross (46.61% of total volume), 96.19% recaptured.
• SOL/USDC (LLTV 62.5%): 8 SVR transactions, $27,790.95 gross (14.59% of total volume), 96.10% recaptured.
• cbLTC/USDC (LLTV 62.5%): 8 SVR transactions, $17,056.41 gross (8.95% of total volume), 95.50% recaptured.
• WETH/USDC (LLTV 86.0%): 9 SVR transactions, $14,907.65 gross (7.82% of total volume), 98.40% recaptured.
• cbADA/USDC (LLTV 62.5%): 11 SVR transactions, $10,488.43 gross (5.51% of total volume), 98.20% recaptured.

Case Study	Tx Hash	Market	Gross ($)	Bid ($)	Recapture	Key Microstructure Finding
Case A: 32-Call Multi-Bundle	0xc0a519...1f9	cbXRP/USDC	$20,969.08	$19,979.88	95.28%	Largest atomic bundle. Solvers liquidated 32 positions simultaneously while returning 95.28% value to depositors.
Case B: Whale Overbid	0x76a4c8...616	cbXRP/USDC	$18,539.32	$19,056.49	**102.79%**	Winner's curse: Intense auction competition forced solver to bid 102.79% of collateral, absorbing a -$522.48 loss.

6. Ethereum L1 MEV-Boost Builder Payment Anatomy

To contextualize L2 SVR efficiency, we analyzed 1,049 verified liquidations on Ethereum L1 across Morpho Blue, Aave v3, Fluid, Spark, and Euler v2. We measured the priority payment transferred from searchers to block builders (`block.coinbase.transfer()`). The data reveals an extreme power-law distribution:
• Micro-Dust Majority: In 848 of 1,049 events (80.84%), builder payments were strictly under $1.00 USD, with a median of $0.08 USD (p25: $0.012, p75: $0.56, p99: $2,639.23).
• Whale Concentration: Total builder payments were $358,316.62 USD. The top 22 transactions accounted for $344,155.62 (96.05%) of all builder revenue. Aave v3 liquidations (133 events) accounted for $317,467.00. Routine liquidations on L1 generate negligible builder competition, whereas whale liquidations route exclusively through private builder bundles.

7. Methodological Scope, Limitations & Reproducibility

Scope Boundaries: (1) Base measurements reflect the acute volatility regime of October 12–19, 2026. (2) Realized solver net profit is lower than reported pre-swap margin due to DEX swap slippage and price impact. (3) L1 measurements quantify MEV-Boost priority payments to builders, not gross collateral volume.
Open Science Reproducibility: The complete JSON dataset (1,157 records) and automated verification scripts are included in this Zenodo archive. Executing `python scripts/verify_dataset_metrics.py` mechanically certifies all 15 core empirical invariants in under two seconds.

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